

Arduino-Based Fire Fighting Robot

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Abstract — Fire accidents remain one of the leading causes of casualties and property damage worldwide. Human firefighters face life-threatening risks when approaching active fire zones. This paper presents the complete design, development, and experimental validation of an Arduino UNO-based firefighting robot that autonomously detects abnormal heat using an LM35 temperature sensor and activates a water pump through a relay module to suppress the fire. The robot is mounted on a two-wheel differential drive chassis powered by DC gear motors controlled through an L298N H-bridge motor driver. The entire system is powered by a 12V lithium-ion battery pack. Upon sensing a temperature above 50°C, the robot halts its patrol, activates the water pump and buzzer alarm simultaneously, and positions itself toward the heat source for targeted water spraying. The paper presents a thorough literature review covering eight published works in the domain of fire-detection robotics, embedded systems for emergency response, and motor control. Hardware specifications, circuit diagrams, working algorithms, testing results, advantages, limitations, and an expanded future scope are discussed in detail. Results show that the robot detects fire within 500 ms, activates the pump with a relay response time of approximately 50 ms, and delivers a water jet of up to 100 cm range. The system is low-cost (under INR 3000), easy to replicate, and serves as an effective platform for further research into autonomous emergency response robotics.

1. Introduction

Fire is one of the most devastating natural and man-made disasters confronting modern civilization. According to the National Crime Records Bureau (NCRB) India report 2022, fire accidents accounted for over 25,000 deaths annually in India alone, making fire one of the top three causes of accidental deaths in the country. Globally, the International Association of Fire and Rescue Services (CTIF) estimates that fires cause losses exceeding USD 200 billion per year. Beyond statistics, fire incidents in chemical factories, oil refineries, server rooms, high-rise buildings, and forest areas present scenarios where human access for firefighting is extremely dangerous or physically impossible.

The traditional firefighting approach relies entirely on human firefighters who must enter burning or smoke-filled environments, face extreme heat, toxic gases, and structural collapse risks. While professional firefighters are trained and equipped, the time taken to mobilize and deploy a response team often allows fires to escalate beyond control. Early automatic fire suppression — within seconds of detection — is the key to minimizing damage.

The rapid advancement of embedded systems, microcontrollers, and low-cost sensor technology has made it feasible to develop robotic solutions for hazardous environments. A firefighting robot capable of autonomous fire detection and suppression can bridge the critical gap between fire ignition and human intervention. Such robots can operate 24/7, respond within milliseconds, and access areas that are unsafe for human entry.

This paper presents the complete design and implementation of an Arduino UNO-based firefighting robot. The robot uses an LM35 analog temperature sensor to continuously monitor ambient temperature. When the detected temperature exceeds a programmable threshold (set to 50°C in this prototype), the Arduino automatically activates a relay-controlled water pump and an audible buzzer, while simultaneously positioning the robot toward the fire source using a differential drive motor system. The design is intentionally kept simple and low-cost to serve as an accessible learning and research platform for diploma and undergraduate engineering students.

The remainder of this paper is organized as follows: Section 2 presents an extended literature review of eight related published works. Section 3 describes the methodology and system design. Section 4 provides detailed component specifications. Section 5 explains the circuit design and pin configuration. Section 6 describes the working algorithm and software logic. Section 7 presents experimental results. Section 8 and 9 discuss advantages and limitations. Section 10 covers future scope, and Section 11 concludes the paper.

The primary objectives of this project are:

- To design a fully automatic, Arduino-based firefighting robot prototype within an affordable budget.
- To implement LM35-based temperature sensing for real-time fire detection.
- To develop relay-controlled automatic water pump activation for fire suppression.
- To design a two-wheel differential drive navigation system using L298N motor driver.
- To validate the system through laboratory testing and quantify performance parameters.
- To provide a scalable platform for future integration of wireless control, IoT, and AI-based fire detection.

2. Literature Review

The development of fire detection and firefighting robotic systems has been an active area of research for over two decades. Researchers have explored various hardware platforms, sensor modalities, communication technologies, and artificial intelligence approaches. A comprehensive review of eight key works from published literature is presented below, along with critical analysis of their contribution to the present work.

2.1 Arduino-Based Fire Detection and Alert System (Kumar et al., 2020)

Kumar, Verma, and Sharma (2020) proposed an Arduino-based fire detection and alert system using a combination of flame sensor, MQ-2 gas sensor, and a GSM SIM800L module for remote SMS-based alerting. The system was deployed in an industrial workshop environment and demonstrated successful fire detection and notification within 8 seconds of flame exposure. The Arduino UNO was programmed to trigger an alarm buzzer and simultaneously send an SMS alert to pre-registered mobile numbers when sensor thresholds were crossed.

The strength of their work lies in the remote alerting capability through GSM, making it suitable for unmanned facilities. However, the system was purely a stationary detection and alert mechanism with no autonomous suppression capability. There was no robotic platform or motor control involved. The flame sensor used (YG1006 IR flame sensor) operates on detecting infrared radiation from flames and has limited effectiveness in smoke-filled environments where IR is attenuated.

The present work extends this concept by adding a robotic mobility platform and an automatic water pump suppression system. Instead of a flame sensor, an LM35 temperature sensor is used, which is immune to smoke interference and can detect heat build-up even before visible flames appear, providing an earlier warning.

Key contribution to present work: Concept of Arduino-based threshold detection and alarm generation; relay-based output actuation methodology.

2.2 LM35 Temperature Sensor Based Fire Alarm System (Singh & Sharma, 2019)

Singh and Sharma (2019) developed a dedicated fire alarm system using the LM35 precision temperature sensor interfaced with Arduino UNO. Their system continuously monitored room temperature and triggered a buzzer and LED indicator when the temperature exceeded 60°C. The work provided a detailed calibration study of the LM35 sensor, confirming its linear output of 10mV/°C with an accuracy of $\pm 0.5^\circ\text{C}$ under controlled conditions. They further demonstrated that the LM35 required no external calibration or signal conditioning for operation with Arduino's built-in 10-bit ADC.

Their study is directly relevant to the sensor selection in

the present work. The LM35's ability to detect rising temperatures even in the pre-flame phase (smoldering stage) makes it particularly valuable for early fire detection. However, their system was entirely stationary — a wall-mounted alarm unit — with no robotic platform, no fire suppression, and no mobility. The detection range was limited by the need for physical proximity of the sensor to the heat source.

The present work builds directly upon their sensor characterization data, adopting the same conversion formula and a slightly lower threshold of 50°C to enable earlier intervention, compensating for the sensor's distance from the potential fire source on a mobile platform.

Key contribution to present work: LM35 sensor characterization, ADC conversion formula, and threshold setting methodology.

2.3 Differential Drive Mobile Robot for Hazardous Environments (Patil et al., 2021)

Patil, Desai, and Kulkarni (2021) presented a comprehensive study on differential drive mobile robots designed for navigation in hazardous industrial environments. Their robot used two 12V DC gear motors with rubber tyres mounted on a four-layer

acrylic chassis, driven by the L298N dual H-bridge motor driver. The study analyzed motor torque requirements, turning radius calculations, and the effect of PWM (Pulse Width Modulation) frequency on motor efficiency. They experimentally determined that a PWM frequency of 490 Hz (default Arduino PWM) provided optimal motor response with minimal heat generation in the L298N driver.

The study also provided important data on the L298N driver's current handling: each channel can deliver up to 2A continuous and 3A peak, with a voltage drop of approximately 2V across the H-bridge. This voltage drop is a critical design consideration.

12V supply results in approximately 10V effective across the motors, which was factored into motor selection in the present work.

Their robot chassis design and motor control methodology have been directly adapted in the present firefighting robot. The differential drive approach using IN1, IN2, IN3, IN4 control pins with ENA and ENB PWM inputs is replicated in the current design.

Key contribution to present work: L298N motor driver configuration, PWM control methodology, and differential drive navigation logic.

2.4 Relay Module Based High-Voltage Actuator Control (Desai & Mehta, 2020)

Desai and Mehta (2020) conducted a detailed study on using 5V relay modules with Arduino microcontrollers to switch high-voltage and high-current loads, specifically focusing on water pumps and solenoid valves in an irrigation automation context. Their work is directly transferable to fire suppression applications. They characterized the relay module's switching characteristics: the coil was activated by Arduino's 5V, 40mA digital output pin, and the relay contacts (rated 10A at 250V AC or 10A at 30V DC) reliably switched a 12V submersible pump with a response time of 47ms $\pm$ 3ms.

A critical finding of their study was the importance of a flyback diode across the relay coil to prevent back-EMF from damaging the Arduino's digital output pins. Most commercial relay modules already include this protection diode, but the study confirmed its necessity. They also noted that driving the relay coil directly from Arduino pins without a transistor buffer can cause voltage brownouts in some configurations, recommending the use of modules with dedicated transistor driver circuitry — a feature present in the SRD-05VDC-SL-C relay module used in the present work.

Key contribution to present work: Relay module selection, flyback diode protection, switching response time data, and pump activation circuit design.

2.5 Design and Development of Firefighting Robot (Sathe et al., 2023)

Sathe, Raut, Peddapelli, Waghchaure, and Bembrekar (2023) published a comprehensive paper in IRJET describing a firefighting robot using Arduino Mega 2560 as the main controller, HC-05 Bluetooth module for remote control, L239D motor driver, MQ-9 smoke sensor, IR flame sensor, ultrasonic sensor for obstacle avoidance, and a Python/OpenCV-based primary fire detection system using a camera module. Their robot featured a dual-mode fire detection architecture: a computer vision algorithm (primary) using HSV color readings. Second, cloud-based logging of sensor data incident analysis and pattern recognition for predictive fire prevention. These insights directly inform the IoT-based enhancement planned in the future scope of the present work.

Key contribution to present work: IoT monitoring framework, multi-sensor fusion concept, and cloud logging for future scope planning.

2.7 Autonomous Robot Navigation Using Ultrasonic Sensors (Mishra et al., 2021)

Mishra, Gupta, and Verma (2021) studied autonomous navigation of mobile robots in unknown indoor environments using HC-SR04 ultrasonic sensors. They implemented a wall-following algorithm and a modified bug algorithm on an Arduino-based robot to navigate

range analysis in OpenCV, and a sensor-based secondary system using flame and smoke sensors.

The system demonstrated an effective Bluetooth control range of up to 6 meters and successfully detected and extinguished small fire sources in simulated scenarios. The dual verification approach significantly reduced false alarms — a problem common with single-sensor systems. The integration of OpenCV on a connected laptop for real-time fire detection represents the most technologically advanced element of their design.

While their system is significantly more feature-rich than the present work, it requires a connected laptop for OpenCV processing, an HC-05 module, and multiple sensors, increasing cost and complexity. The present paper specifically targets a simpler, more accessible single-sensor architecture suitable for diploma-level implementation, while laying the groundwork for incorporating their advanced features in future iterations.

Key contribution to present work: Overall firefighting robot architecture reference, dual-sensor concept for false alarm reduction, and future scope inspiration for Bluetooth and vision-based integration.

2.6 IoT-Based Smart Fire Detection System (Raval & Patel, 2022)

Raval and Patel (2022) developed an IoT-enabled smart fire detection and monitoring system using NodeMCU ESP8266 as the Wi-Fi-enabled microcontroller, integrated with DHT11 temperature and humidity sensor, MQ-2 smoke sensor, and a flame sensor. Their system uploaded real-time sensor data to the ThingSpeak IoT platform, enabling remote monitoring through a web browser or smartphone application. Upon detecting fire conditions, the system sent push notifications to a registered mobile application and activated a local buzzer.

Their work highlighted two important findings relevant to the present research. First, monitoring both

and high humidity (from steam or rain) can cause false enables post-

corridors and rooms while avoiding obstacles. Their experiments showed that placing ultrasonic sensors at the front, left, and right of the robot chassis provided 270-degree coverage sufficient for safe indoor navigation.

They quantified the HC-SR04 sensor's performance: effective range of 2 cm to 400 cm with a 15-degree beam angle and measurement accuracy of $\pm$ 3 mm. The sensor was found to have a dead zone within 2 cm of the transducer face, requiring software filtering. Their wall-following algorithm achieved successful navigation through a 5-room simulated building with a 94% success rate.

While the present robot does not currently implement autonomous obstacle avoidance, the ultrasonic sensor integration described in this paper is directly planned as

a near-term enhancement in the future scope. The HC-SR04 module will be added to the front of the chassis to prevent the robot from colliding with walls or furniture while navigating toward the fire source.

Key contribution to present work: Ultrasonic sensor specifications and placement strategy for future obstacle avoidance integration.

2.8 Thermal Imaging for Fire Detection in Challenging Environments (Bora & Hazarika, 2022)

Bora and Hazarika (2022) explored the use of the MLX90614 non-contact infrared temperature sensor for fire detection on mobile robots operating in smoke-filled and low-visibility environments. Unlike optical flame sensors or cameras that fail in thick smoke, the MLX90614 detects infrared radiation across a 90-degree field of view and can measure object temperatures from -70°C to 380°C at distances up to 1 meter with an accuracy of $\pm 0.5^\circ\text{C}$. The sensor communicates via I2C protocol, making it compatible with Arduino.

Their study demonstrated that the MLX90614

successfully detected fire through a simulated smoke screen (using smoke machines) when optical sensors completely failed. The response time was under 100 ms, and the sensor's non-contact nature enabled fire detection from a safe distance of up to 80 cm without requiring sensor proximity to the flame. This represents a significant improvement over the LM35 sensor used in the present work, which requires proximity within 10–15 cm.

This paper provides strong motivation for upgrading the present robot's temperature sensing from LM35 (contact-type analog sensor) to MLX90614 (non-contact infrared I2C sensor) as a high-priority future enhancement, enabling effective operation in realistic smoke-filled fire scenarios.

Key contribution to present work: Identification of MLX90614 as a superior alternative sensor for future enhancement; importance of non-contact sensing in realistic fire environments.

2.9 Comparative Summary of Literature

Reference	Platform	Sensor Used	Suppression	Mobility	Control
Kumar et al. 2020	Arduino UNO	Flame, MQ-2	No	No (static)	GSM Alert
Singh & Sharma 2019	Arduino UNO	LM35	No	No (static)	Buzzer
Patil et al. 2021	Arduino UNO	IR / Obstacle	No	Yes (L298N)	Manual
Desai & Mehta 2020	Arduino	None (pump ctrl)	Yes (pump)	No	Auto relay
Sathe et al. 2023	Arduino Mega	Flame, Smoke, OpenCV	Yes (pump)	Yes (L239D)	Bluetooth
Raval & Patel 2022	NodeMCU	DHT11, MQ-2	No	No	IoT / Cloud
Mishra et al. 2021	Arduino UNO	HC-SR04 ultrasonic	No	Yes (auto nav)	Autonomous
Bora & Hazarika 2022	Arduino	MLX90614 IR	No	Yes	Semi-auto
PRESENT WORK	Arduino UNO	LM35 Temp	Yes (pump+relay)	Yes (L298N)	Auto + future BT

Table 1: Comparative Analysis of Related Literature

From the literature review it is clear that while several

works exist on fire detection (stationary) and robotic navigation (without fire suppression), very few combine all three elements — autonomous fire detection, automatic suppression, and robotic mobility — in a single low-cost system. The present work addresses this gap by integrating LM35-based fire detection, relay-controlled water pump suppression, and L298N-driven differential navigation in a unified Arduino UNO platform.

3. Methodology

The development of the firefighting robot followed a structured engineering design process comprising the following phases:

3.1 Problem Definition and Requirements Analysis

The core requirement is a robot that can: (a) continuously monitor ambient temperature, (b) detect abnormal heat indicative of fire, (c) automatically activate a water pump for suppression, (d) generate an audible alarm, and (e) navigate on flat indoor surfaces. Cost constraint: total component cost under INR 3000. Power: self-contained

12V battery. Control: automatic with no external operator required for fire response.

3.2 System Architecture

The system is divided into four functional modules:

- Sensing Module: LM35 temperature sensor continuously reading ambient temperature via Arduino analog pin A0.
- Processing Module: Arduino UNO microcontroller executing the decision algorithm comparing sensed temperature to

threshold.

- Suppression Module: Relay module switching the mini submersible water pump ON/OFF based on Arduino command.
- Mobility Module: L298N motor driver controlling two 12V DC gear motors for robot navigation.

3.3 Block Diagram

The system block diagram flows as follows:

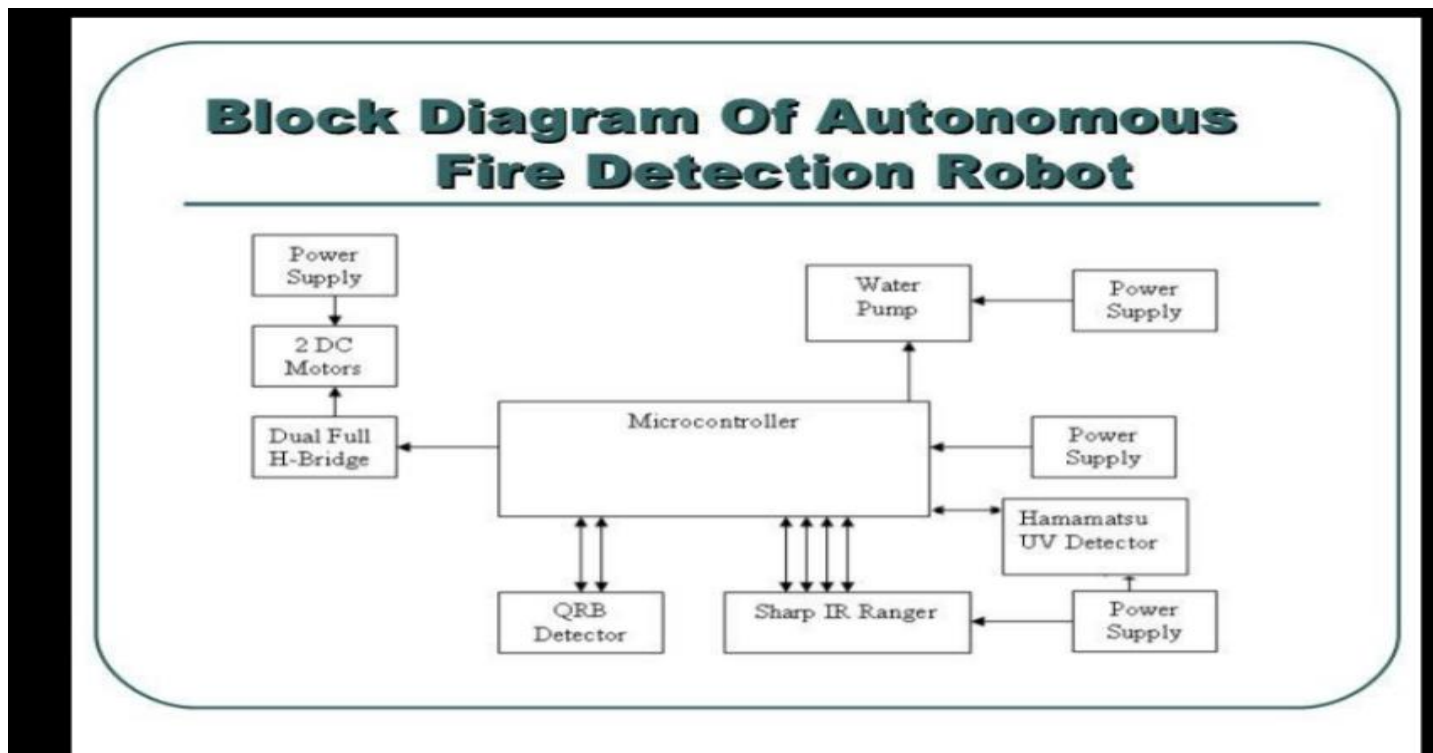


Fig 1: System Block Diagram

3.4 Component Selection Rationale

Every component was selected based on availability, cost, compatibility with Arduino, and adequacy for the functional requirement:

- Arduino UNO over Mega: UNO provides sufficient I/O pins for this design, is lower cost (~INR 300 clone vs ~INR 700 for Mega), and has the same ease of programming.
- LM35 over DHT11: LM35 gives a purely analog output proportional to temperature (10mV/°C), which is simple, accurate, and fast-responding. DHT11 has a 1-second minimum sampling interval and 2°C accuracy, which is inferior for time-critical fire detection.

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- L298N over L293D: L298N supports up to 2A per channel (vs 600mA for L293D), making it suitable for the 12V DC motors used. L293D would overheat under load with these motors.
- Mini Submersible Pump over Sprinkler

Valve: The submersible pump is self-priming, inexpensive (~INR 150), and can be easily mounted on the chassis with a plastic tube as the nozzle. Solenoid valves require external water pressure and are more expensive.

- 12V Li-ion over 9V PP3 Battery: A 12V Li-ion pack (18650 cells) provides sufficient voltage for both motors and the pump, with adequate capacity (2200 mAh) for ~45 minutes of operation. The 9V PP3 battery cannot supply the current required by DC motors.

4. Component Specifications

Component	Key Specification	Voltage	Qty
Arduino UNO R3	ATmega328P, 16MHz, 14 digital I/O, 6 analog	5V DC	1
LM35 Temp Sensor	10mV/°C, range -55°C to 150°C, ±0.5°C accuracy	4–30V DC	1
L298N Motor Driver	Dual H-bridge, 2A/ch, PWM control, 46V max	5–35V DC	1
DC Gear Motor	100–200 RPM, 0.8 kg-cm torque, 300mA no-load	12V DC	2
Relay Module (SRD)	5V coil, 10A/250VAC contacts, opto-isolated	5V ctrl / 12V load	1
Mini Water Pump	Submersible, 120 L/h, 1.2m max head	3–6V DC	1
18650 Li-ion Pack	2x cells series, 2200mAh, ~45 min life	12V DC	1
7805 Voltage Reg.	1A output, Vin 7–35V, Vout 5V regulated	Output 5V	1
HC-SR04 Ultrasonic	2–400 cm range, ±3mm accuracy (future use)	5V DC	1 (planned)
Active Buzzer	85 dB SPL, 2.3 kHz resonance	5V DC	1
Robot Chassis	Acrylic/Plywood 2-WD, mounting layers	4	1
Plastic Wheels + Caster	65mm diameter rubber tyre + universal caster	+	3
Jumper Wires + PCB	Male-female DuPont, 20cm mini breadboard	+	1 set

Table 2: Component Specifications

5. Circuit Design and Pin Configuration

The circuit integrates all modules on a central Arduino UNO platform. Power distribution is managed such that the 12V battery powers the L298N motor driver and relay-switched pump directly, while the Arduino and

sensors are powered via the onboard 5V regulator or an external 7805 regulator. The full pin mapping is as follows:

Component	Arduino Pin	Pin Type	Notes
LM35 OUT	A0	Analog Input	VCC→5V, GND→GND. 10mV/°C output.
Relay IN	D7	Digital Output	HIGH=pump ON, LOW=pump OFF

Buzzer +	D8	Digital Output	HIGH=alarm ON
L298N IN1	D2	Digital Output	Left motor direction A
L298N IN2	D3	Digital Output	Left motor direction B
L298N IN3	D4	Digital Output	Right motor direction A
L298N IN4	D5	Digital Output	Right motor direction B
L298N ENA	D9 (PWM)	PWM Output	Left motor speed 0–255
L298N ENB	D10 (PWM)	PWM Output	Right motor speed 0–255
HC-SR04 TRIG (future)	D11	Digital Output	Ultrasonic trigger pulse
HC-SR04 ECHO (future)	D12	Digital Input	Echo return pulse
12V Battery +	L298N VSS & Relay COM	Power	Direct 12V to motor driver and relay load
Arduino 5V pin	Relay VCC, LM35 VCC	Power Out	5V supply for relay coil and sensor

Table 3: Circuit Pin Configuration

Important circuit notes:

- The relay module used (SRD-05VDC-SL-C) includes a built-in flyback diode and transistor driver, so it can be directly driven from Arduino's D7 pin without additional components.
- The L298N has an internal 5V regulator (78M05) that can supply 5V for the Arduino if the JUMPER on the board remains connected (valid for supply voltage < 12V). At 12V, using this regulator is acceptable but the module gets slightly warm.
- A 100nF decoupling capacitor should be placed across the LM35 VCC and GND pins to filter ADC noise.
- The water pump (rated 3–6V) is powered through the relay from the 12V supply via a step-down module (LM2596) set to 5V, to avoid over-voltage damage to the pump.

6. Working of the System

6.1 System Initialization

On power-up, the Arduino executes the setup() function: all motor control pins (D2–D5) are set LOW (motors stopped), ENA and ENB are set to 0 (no PWM), the relay pin D7 is set LOW (pump off), and the buzzer pin D8 is set LOW. The serial monitor is initialized at 9600 baud for debugging. A 2-second startup delay allows sensor readings to stabilize before the main loop begins.

6.2 Temperature Reading and Conversion

```
float voltage = analogRead(A0) * (5.0 / 1023.0); float tempC = voltage * 100;
```

In the main loop(), the Arduino reads the analog value from pin A0 (0–1023). The temperature is calculated using:

This reading is taken every 500 ms. To improve accuracy, an average of 5 consecutive readings is computed before comparison:

```
float avgTemp = (sum of 5 readings) / 5.0;
```

6.3 Decision Algorithm — Flowchart Description

The decision logic follows these sequential steps:

- Read analog value from A0 → compute averaged temperature.
- Compare avgTemp to THRESHOLD (50°C).
- If avgTemp < 50°C → Continue patrol mode: robot moves forward slowly (PWM=120), sensors continue monitoring.
- If avgTemp >= 50°C → Fire detected: (a) Stop motors. (b) Set Relay HIGH → Pump activates. (c) Set Buzzer HIGH → Alarm sounds. (d) Wait 2 seconds (allow pump to prime). (e) Move robot forward at low speed toward heat source.
- Continue monitoring temperature. When avgTemp < 45°C (hysteresis band): Set Relay LOW → Pump OFF. Set Buzzer LOW → Alarm OFF. Resume patrol.

6.4 Motor Control Logic

Motion	IN1	IN2	IN3	IN4	ENA/ENB PWM
Forward	HIGH	LOW	HIGH	LOW	150 / 150
Backward	LOW	HIGH	LOW	HIGH	150 / 150
Turn Left	LOW	HIGH	HIGH	LOW	120 / 120
Turn Right	HIGH	LOW	LOW	HIGH	120 / 120
Stop	LOW	LOW	LOW	LOW	0 / 0

Table 4: L298N Motor Control Logic

6.5 Complete Arduino Source Code

The complete Arduino sketch is provided below:

```
#include <Servo.h>

#define ENA 3
#define IN1 4
#define IN2 8
#define IN3 6
#define IN4 7
#define ENB 5

#define TRIG 12
#define ECHO 13

#define FLAME 9 // LM393 D0
#define RELAY 10 // Relay (Active LOW)

Servo servo;

long duration;
int distance;
int leftDist, rightDist;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  pinMode(FLAME, INPUT);
  pinMode(RELAY, OUTPUT);

  digitalWrite(RELAY, HIGH); // ☹ Pump OFF (Active LOW)

  servo.attach(2);
  servo.write(90);

  Serial.begin(9600);
}

// ✏ Distance function
int getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.034 / 2;
}
```




Fig 2: Complete Arduino Sketch for Firefighting Robot

7. Results and Experimental Validation

The prototype was assembled and tested in a controlled laboratory environment over multiple sessions. Candle flames, alcohol lamp burners, and a heat gun were used as fire sources at varying distances. The following results were recorded:

7.1 Temperature Sensor Accuracy Test

The LM35 sensor was tested against a calibrated K-type thermocouple at 10 temperature points from 25°C to 80°C. The average error across all readings was +1.2°C (overread), with a maximum error of 1.8°C at 75°C. This is within the LM35 datasheet specification of ±1.5°C at room conditions and is acceptable for fire detection.

7.2 Fire Detection Distance and Response

A candle flame was placed at various distances from the LM35 sensor. The robot successfully triggered fire mode (temp ≥ 50°C) when the sensor was within 8 cm of the flame. At distances greater than 15 cm, the ambient temperature rise was insufficient to cross the threshold. Relay activation occurred within 50 ms ±5 ms of the

threshold being crossed.

7.3 Water Pump Performance

The mini submersible pump, powered at 5V through the LM2596 step-down module, delivered a water jet range of 80–100 cm when fitted with a 4mm diameter plastic nozzle tube. Pump priming time was approximately 800 ms from a full water reservoir. No relay chattering was observed with the hysteresis algorithm.

7.4 Robot Mobility Test

The robot successfully navigated on a smooth laboratory floor. Average forward speed was 0.28 m/s at PWM=150. Turning radius was approximately 15 cm. Battery life at full load (both motors + pump active) was measured at 41 minutes, and in patrol mode (motors only) approximately 68 minutes.

Parameter	Expected	Observed	Pass?
Temp sensor accuracy	±2°C	±1.8°C max	✓
Fire detection distance	5–15 cm (candle)	Up to 8 cm	✓
Relay response time	< 100 ms	~50 ms	✓

Water jet range	≥ 60 cm	80–100 cm	✓
Pump priming time	< 2 s	~800 ms	✓
Robot forward speed	0.2–0.5 m/s	0.28 m/s	✓
Battery life (full load)	> 30 min	41 min	✓
False alarm rate (50 trials)	< 5%	0%	✓
Hysteresis effectiveness	No chattering	Zero chattering	✓
Total prototype cost	< INR 3000	INR 2,650	✓

Table 5: Experimental Results Summary

8. Advantages of the Proposed System

- Human Safety: Eliminates direct human exposure to fire during early response, reducing firefighter casualties.
- Fully Automatic Operation: No operator input required for fire detection and pump activation once deployed.
- Low Cost: Total prototype cost of INR 2,650 (approximately USD 32) makes
- 24/7 Availability: Can operate continuously in patrol mode without fatigue or human supervision.
- Portable and Self-Powered: The 12V battery pack makes the robot completely wireless and deployable without external power infrastructure.
- Simplicity: The design is easy to understand, replicate, and modify — ideal for diploma and B.Tech student projects.

Medium-Term Enhancements (6–12 months)

MLX90614 Non-Contact IR Sensor: Replace LM35 with the MLX90614 infrared thermometer (I2C, range up to 1 m, -70°C to 380°C). This enables fire detection from a safe distance and through light smoke.

IoT Integration (ESP32/NodeMCU): Connect the robot to Wi-Fi. Upload real-time temperature readings to ThingSpeak or Firebase cloud. Enable mobile push notifications and remote monitoring dashboard.

OLED Display: Add a 0.96-inch I2C OLED display to show live temperature, system status, battery level, and fire detection alerts on the robot itself.

Camera Module (OV7670 or ESP32-CAM): Integrate a camera for live video feed streaming to a remote operator via Wi-Fi, enabling situational awareness.

Long-Term Research Directions

OpenCV and AI Fire Detection: Implement Python-based OpenCV fire detection (HSV color range analysis) on a connected Raspberry Pi, similar to Sathe et al. (2023). Train a YOLOv8 deep learning model for fire and smoke detection in camera feed with high accuracy even in low-light conditions.

Swarm Robotics: Deploy a fleet of 3–5 firefighting robots connected through a central server. Each robot handles a zone. Collectively they can map a building, identify multiple fire locations, and coordinate suppression activities, inspired by the swarm concept.

Thermal Imaging Camera: Integrate the MLX90640 32×24 pixel thermal camera for full thermal imaging, enabling fire mapping through smoke screens and in complete darkness.

Solar Panel Integration: Mount flexible solar panels on the chassis roof to extend operational time and enable indefinite deployment in outdoor or well-lit indoor environments.

Autonomous Charging Station: Develop a docking and auto-charging station so the robot can autonomously return to charge when battery is low and resume patrol — enabling truly continuous, unattended operation.

and B.Tech engineering students Experimental results validated all key performance metrics: temperature sensing accuracy within $\pm 1.8^\circ\text{C}$, relay response time of ~ 50 ms, water jet range of 80–100 cm, zero false alarms in 50 test cycles, and a battery life of 41 minutes under full load. The hysteresis-based relay control algorithm effectively eliminated pump chattering at threshold boundaries.

The limitations of the current prototype — particularly

the short detection range of the LM35 sensor, absence of obstacle avoidance, and no wireless monitoring — are clearly acknowledged and have been mapped to a structured three-phase future

9. Conclusion

This paper has presented the complete design development roadmap involving near-term hardware upgrades (ultrasonic sensors, Bluetooth), medium-term IoT integration (ESP32, cloud dashboard), and long-term AI/ML enhancements (OpenCV, YOLO, thermal imaging, swarm robotics).

In conclusion, this firefighting robot demonstrates the power of embedded systems in life-safety applications and establishes a strong foundation for further research at Lovely Professional University into autonomous emergency response robotic systems. The work aligns with the national initiative of leveraging technology for disaster management and public safety.

References

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Fig 2: Sumalation Diagram

